JAYA SRAVAN KALA

Computer Vision / Machine Learning Engineer

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PROFESSIONAL SUMMARY

Computer Vision and ML Engineer with 3.5+ years of experience: ~2 years building production computer vision systems at Sam's Club, plus large-scale Spark and Kafka data engineering. Builds end-to-end systems that combine YOLO detection and segmentation, DOLG embeddings and Gemini VLM verification to monitor inventory across ~500 store cameras, with a focus on precision, reducing false positives and production reliability.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Bash
Computer Vision & ML: PyTorch, TorchScript, YOLO / YOLOv11 (object detection, instance segmentation, OBB), Ultralytics, OpenCV, DOLG embeddings (image retrieval), Grounding DINO, EasyOCR, PaddleOCR, scikit-learn, Optuna, CLAHE, GPU inference (CUDA)
Vision-Language & GenAI: Gemini (VLM), prompt engineering, structured JSON outputs, CLIP, ViLT
Geometry & Matching: Shapely (polygon IoU), homography projection, RapidFuzz fuzzy matching
MLOps & Evaluation: MLflow Model Registry, model evaluation (mAP, precision/recall, TP/FP analysis), pseudo-labeling, Docker, CI/CD, Git, PyTest, Jupyter
Cloud: Azure (Cosmos DB, Blob, Service Bus, Data Factory), GCP (BigQuery, GCS, Dataproc), AWS S3, service principal auth (Akeyless)
Data Engineering: Apache Spark (PySpark, Scala, Structured Streaming), Apache Kafka, Airflow, Delta Lake, Cassandra, Pandas, NumPy, Snowflake, MySQL, Talend, Tableau, Excel
Engineering: Multithreading and concurrency, REST APIs, Linux, SRE runbooks, root-cause analysis

PROFESSIONAL EXPERIENCE

Sam's Club | *Computer Vision Engineer* | *Sunnyvale, CA* **Feb 2025 – Present**

- Built production computer vision alerting across ~500 store cameras that detects product changes, unmapped products and restocks several times a day, publishing results to Cosmos DB, BigQuery and Kafka.
- Cut alerts sent to human review by 46% (310 to 168 in a live run) with a two-stage gate: a DOLG embedding prefilter followed by Gemini VLM verification that filters out shadows, lighting shifts and restocking noise.
- Filtered out 66% of reviewed restock and depletion alerts as routine restocks by re-checking each shelf bin after 30 and 60 minutes; shipped to production.
- Matched bins across overlapping cameras using floor-plan homography and Shapely polygon IoU to confirm real changes, share verified alerts and suppress duplicates.
- Trained YOLOv11 detection, instance segmentation and OBB models in PyTorch with Optuna tuning, reaching 97% mAP@50 on key product classes; versioned models in MLflow and ran GPU-batched TorchScript inference.
- Developed a price-tag mismatch detector (YOLO + multi-pass EasyOCR/PaddleOCR with CLAHE and fuzzy catalog matching), reaching 100% tag-detection recall and 97.1% OCR recall on validation; its output feeds a UPC change recommendation pipeline.
- Created a QA alert pipeline (six alert types) that flags likely mislabeled products across 160 cameras by comparing each day's image with 14 days of per-product DOLG templates; demoed it to stakeholders.
- Engineered a misassigned-UPC detector that checks each bin's product against its catalog image and description with a VLM, confirms mismatches in a second pass and makes the final call with deterministic rules; added caching and retries.
- Consolidated three inventory tracking models into one segmentation model and migrated the downstream schema.
- Designed segmentation post-processing (bin splitting, NMS, mask-IoU filtering) that removes duplicate and overlapping detections on live camera feeds.
- Automated image ingestion from enterprise camera APIs with quality checks (Laplacian blur filtering, fisheye correction) and Azure Service Bus pseudo-labeling to reduce manual annotation.
- Prototyped high-resolution GTIN verification using Grounding DINO with tiled inference on 4K images, multi-orientation OCR and check-digit correction to flag product barcode mismatches.
- Delivered a daily report that finds sold products missing from the store's item-location map, joining sales receipts, the location map and the product catalog with UPC normalization, item-number roll-up and fuzzy name matching.
- Set up TP/FP dashboards, an accuracy validator and a DOLG+VLM vs. VLM-only comparison to measure alert precision, plus HTML review tools for labelers and threshold calibration.
- Hardened production code: fixed static-analysis findings (remote-code-execution risks, Cosmos DB SQL injection), moved Cosmos DB to service principal auth via Akeyless, wrote multi-stage Dockerfiles with PyTest coverage, and authored SRE runbooks.

UCode Technologies LLC | *Software Trainee* | *Ashburn, VA* **Sep 2024 – Jan 2025**

- Developed and maintained components of the Scala/Spark core engine of a metadata-driven data platform processing 100M+ records per day from Kafka, Cassandra, GCS and REST API sources.
- Designed pluggable source-to-GCP connectors so teams could onboard new sources through configuration instead of code.

- Delivered live ad-event data to model training with no lost or duplicate events by building Kafka-to-GCS Spark Structured Streaming jobs with manual offset commits and SSL/TLS.
- Implemented micro-batch and near-real-time Kafka pipelines for merchandising and operations teams.
- Wrote and validated Spark transformations feeding replenishment, pricing, forecasting and personalization models.
- Reduced Spark runtimes by up to 80% (one supply-chain job from 6–8 hours to under 1 hour) using Adaptive Query Execution, partitioning, caching, broadcast joins and key salting.
- Orchestrated pipelines for multiple business units in Airflow with tenant-based routing and dev/stage/prod configs; set up Airflow CI/CD with Docker-based deployments to Astronomer.

Cognizant | *Program Analyst* **Feb 2021 – Jul 2022**

- Ran and monitored daily, weekly and monthly ETL jobs in Talend, extracting data from multiple source systems and loading it into MySQL and Snowflake on schedule and within SLA.
- Transformed and cleaned data with SQL in Snowflake and Excel (formulas, lookups), and delivered curated datasets to Tableau dashboards, working with the dashboard team to fix issues.
- Worked directly with the client to gather reporting requirements and updated jobs and transformations as needs changed.

PROJECTS

BEV 3D Perception & Tracking on nuScenes | *BEVDet*, *Kalman filter*, *ONNX*, *TensorRT*

- Building a camera-based bird's-eye-view 3D detection and multi-object tracking pipeline on nuScenes (6 surround cameras), evaluated with mAP, NDS and AMOTA.
- Adding tracking-by-detection with Hungarian matching and a Kalman filter, and benchmarking FP32/FP16/INT8 TensorRT inference for latency vs. accuracy.

Multimodal Fake News Detection | *CLIP*, *ViLT*

- Developed a multimodal detector that uses CLIP and ViLT embeddings to catch mismatches between article text and images.
- Scored text-image consistency with cosine similarity to flag fabricated context, with attention maps to explain results.

Formula One Data Pipeline | *Azure Data Factory*, *PySpark*, *Delta Lake*

- Built an Azure pipeline that ingests live race telemetry and historical driver statistics from REST APIs into Azure Data Lake.
- Added schema validation and partition optimization to speed up analytical queries for downstream dashboards.

EDUCATION

George Mason University – Master of Science, Computer Science | Fairfax, VA Aug 2022 – May 2024

Institute of Aeronautical Engineering – B.Tech, Information Technology | Hyderabad, India Aug 2017 – May 2021

CERTIFICATIONS

AWS Certified Cloud Practitioner | Microsoft Certified: Azure Data Engineer Associate (DP-203)